# Comprehensive Practice Test (Units 1 - 5)

## Section 1: Multiple Choice Questions

*(No calculator is allowed for these questions. )*

**1.** What is the value of $\lim_{x \to 2} \frac{x^2-4}{e^{x-2}-1}$ ?

(A) $0$

(B) $1$

(C) $4$

(D) The limit does not exist.

**2.** Let $f$ be the function defined below, where $k$ is a constant.

$$f(x) = \begin{cases} \frac{\sin(kx)}{x}, & \text{for } x < 0 \\ 3x + k^2, & \text{for } x \geq 0 \end{cases}$$

For what value(s) of $k$ is $f$ continuous at $x = 0$ ?

(A) $k = 0$ only

(B) $k = 1$ only

(C) $k = 0$ and $k = 1$

(D) There is no such value of $k$.

**3.** $\lim_{h \to 0} \frac{\cos\left(\frac{\pi}{4}+h\right)-\frac{\sqrt{2}}{2}}{h} =$

(A) $-\frac{\sqrt{2}}{2}$

(B) $0$

(C) $\frac{\sqrt{2}}{2}$

(D) The limit does not exist.

**4.** The table below gives values of the differentiable functions $f$ and $g$ and their derivatives at $x = 2$ and $x = 3$.

| $x$ | $f(x)$ | $f'(x)$ | $g(x)$ | $g'(x)$ |
|---|---|---|---|---|
| $2$ | $4$ | $5$ | $3$ | $4$ |
| $3$ | $1$ | $-2$ | $2$ | $6$ |

If $h(x) = f(g(x))$, what is the value of $h'(2)$ ?

(A) $-8$

(B) $20$

(C) $24$

(D) $26$

**5.** If $x^2 + xy + y^2 = 7$, what is the value of $\frac{dy}{dx}$ at the point $(1, 2)$ ?

(A) $-1$

(B) $-\frac{4}{5}$

(C) $-\frac{2}{3}$

(D) $\frac{4}{5}$

**6.** Let $f(x) = x^3 + 2x - 1$, and let $g(x) = f^{-1}(x)$. What is the value of $g'(2)$ ?

(A) $\frac{1}{14}$

(B) $\frac{1}{5}$

(C) $5$

(D) $14$

**7.** The radius of a circle is increasing at a constant rate of $\frac{2}{\pi}$ centimeters per second. What is the rate of change of the area of the circle, in square centimeters per second, at the instant when the radius is $5$ centimeters?

(A) $10$

(B) $20$

(C) $10\pi$

(D) $20\pi$

**8.** A particle moves along the x-axis so that its position at time $t$ is given by $x(t) = t^3 - 6t^2 + 9t + 1$. For what values of $t$ is the particle moving to the left?

(A) $t < 1$

(B) $t > 3$

(C) $1 < t < 3$

(D) $0 < t < 2$

**9.** The function $f$ is given by $f(x) = x^3 - 3x$. The Mean Value Theorem guarantees that there exists a number $c$ in the open interval $(0, 2)$ such that $f'(c)$ is equal to the average rate of change of $f$ on the closed interval $[0, 2]$. What is the value of $c$ ?

(A) $\frac{2}{\sqrt{3}}$

(B) $\sqrt{2}$

(C) $\frac{4}{3}$

(D) $1$

**10.** Let $f$ be a polynomial function with derivative given by $f'(x) = (x - 2)^2(x + 1)$. Which of the following statements is true about the relative extrema of $f$ ?

(A) $f$ has a relative maximum at $x = -1$ and a relative minimum at $x = 2$.

(B) $f$ has a relative minimum at $x = -1$ and a relative maximum at $x = 2$.

(C) $f$ has a relative minimum at $x = -1$ and no relative extremum at $x = 2$.

(D) $f$ has a relative maximum at $x = 2$ and no relative extremum at $x = -1$.

**11.** Let $f$ be a function whose second derivative is given by $f''(x) = x(x - 3)^3(x + 2)^2$. How many points of inflection does the graph of $f$ have?

(A) One

(B) Two

(C) Three

(D) Four

**12.** Let $f$ be a twice-differentiable function. If $f'(x)$ has a relative maximum at $x = a$, which of the following must be true?

(A) $f(x)$ has a relative maximum at $x = a$.

(B) $f(x)$ has a relative minimum at $x = a$.

(C) The graph of $f(x)$ has a point of inflection at $x = a$.

(D) $f''(a) > 0$.

**13.** Let $f$ and $g$ be differentiable functions such that $\lim_{x \to 0} \frac{f(x) - 2}{g(x)} = 3$. If $g(0) = 0$, what must be the value of $f(0)$ ?

(A) $0$

(B) $2$

(C) $3$

(D) $5$

**14.** If $y = \ln(3x)$, what is $\frac{d^2y}{dx^2}$ ?

(A) $-\frac{1}{x^2}$

(B) $-\frac{1}{3x^2}$

(C) $\frac{1}{x^2}$

(D) $-\frac{9}{x^2}$

**15.** A rectangle has a fixed perimeter of $20$ units. What is the maximum possible area of this rectangle?

(A) $20$

(B) $24$

(C) $25$

(D) $100$

# Section 2: Free Response Questions

*(Show all your work clearly. Your work must be expressed in proper mathematical notation. No calculator is allowed.)*

## Question 1

A particle moves along the x-axis so that its velocity $v$ at time $t$ is given by $v(t) = t^3 - 6t^2 + 8t$ for $0 \leq t \leq 5$.

**(a)** Find the acceleration $a(t)$ of the particle at time $t$. Is the speed of the particle increasing or decreasing at time $t = 3$ ? Give a reason for your answer.

**(b)** Find all times $t$ in the open interval $0 < t < 5$ at which the particle changes direction. Justify your answer.

**(c)** Find the maximum acceleration of the particle on the closed interval $[0, 5]$. Justify your answer.

## Question 2

Consider the curve defined by the equation $x^2 + xy + y^2 = 27$.

**(a)** Show that $\frac{dy}{dx} = \frac{-2x-y}{x+2y}$.

**(b)** Write an equation for the line tangent to the curve at the point $(3, 3)$.

**(c)** Find the x-coordinates of all points on the curve where the tangent line is vertical.

## Question 3

Let $f$ be the function defined by $f(x) = x^4 - 4x^3 + 10$.

**(a)** Find the x-coordinates of all relative extrema of $f$. Classify each as a relative maximum or a relative minimum. Justify your answers.

**(b)** Find the x-coordinates of all points of inflection of the graph of $f$. Justify your answer.

**(c)** Find the absolute minimum value of $f(x)$ on the closed interval $[-1, 4]$.

## Question 4

A ladder $13$ feet long is leaning against a vertical wall. The bottom of the ladder is sliding away from the base of the wall at a constant rate of $2$ feet per second.

**(a)** How fast is the top of the ladder sliding down the wall at the instant when the bottom of the ladder is $5$ feet from the wall?

**(b)** At what rate is the area of the right triangle formed by the ladder, the wall, and the ground changing at the instant when the bottom of the ladder is $5$ feet from the wall? Include units in your answer.

**(c)** Let $\theta$ be the angle between the ladder and the ground. At what rate is the angle $\theta$ changing at the instant when the bottom of the ladder is $5$ feet from the wall?